

SEMESTER-V

Category I

(B.A. Honours in Geography in three years)

DISCIPLINE SPECIFIC CORE COURSE – ENVIRONMENT AND ECOLOGY (DSC 13)

Course title & Code	Credits	Duration (Hrs per week)			Eligibility Criteria	Prerequisite
		Lecture	Tutorial	Practical/ Practice		
ENVIRONMENT AND ECOLOGY	4	3	1	0	Class 12th	NIL

Learning Objectives:

1. Various dimensions of ecology and ecosystems, their spatial distribution.
2. To learn about the global environmental challenges and management
3. To know about regional environmental challenges.
4. Understanding of environmental governance.

Learning Outcomes:

1. Detailed exposure to the concept of ecology, ecosystem, processes, theories and concepts.
2. In-depth knowledge of anthropogenic interventions and impacts, conservation strategies and planning.
3. Understanding the environmental concerns at global and regional level.
4. Evaluation and achievement of different environmental programs, policies and legislations.

Course Outline:

Unit-1 Introduction: (7 hrs)

- Concept of Environment, Ecology and Ecosystem; Types of Ecology; Concepts of Ecosystem Services; Ecological and Material Footprint; Global Planetary Boundaries.

Unit-2 Ecology and Ecosystem: (9 hrs)

- Species Interactions; Ecological Limiting Factors; Ecosystem: Structure and Functions; Human Adaptation

Unit-3 Global Environmental Challenges and Management: (11 hrs)

- Climate Change, Biodiversity loss, Land degradation and Human health issues

Unit-4 Regional Ecological Issues and Management: (11 hrs)

- Coastal and Marine Ecology: Loss of mangroves and corals, Garbage Patches; Urban Ecology: Waste disposal and Pollution

Unit-5 Programmes and Policies: (7 hrs)

- Environmental Impact Assessment; Global and National Environment Policy of India

Readings:

- Brewster, E. N. 2010. Climate Change Adaptation: Steps for a Vulnerable Planet, New York, Nova Science.
- Cain, M.L., Bowman, W.D. and Hacker S.D. (2011). Ecology, 2nd Edition, Sinauer Associates Inc.
- Chandna R. C., 2002: *Environmental Geography*, Kalyani, Ludhiana.
- Chapman, J.L. & M.J. Reiss. (1998). Ecology: Principles and Applications. Cambridge Univ. press.
- Cunningham W. P. and Cunningham M. A., 2004: *Principals of Environmental Science: Inquiry and Applications*, Tata Macgraw Hill, New Delhi.
- Das, R. C., 1998. The Environmental Divide: The Dilemma of Developing Countries, A.P.H. Pub., New Delhi.
- Freedman, Bill. 1995. Environmental Ecology: The Ecological Effects of Pollution, Disturbance, and Other Stresses, Academic Press. London.
- Global Environment Monitoring UNEP, <https://wesr.unep.org/article/global-environment-monitoring>
- Global Environmental Outlook Reports UNEP <https://www.unep.org/geo/>
Intergovernmental Panel on Climate Change IPCC Reports(2021-23)
<https://www.ipcc.ch/report/ar6/wg2/>

**DISCIPLINE SPECIFIC CORE COURSE – AGRICULTURAL
GEOGRAPHY AND FOOD SECURITY (DSC 14)**

Course title & Code	Credits	Duration (per week)			Eligibility Criteria	Prerequisite
		Lecture	Tutorial	Practical/ Practice		
AGRICULTURAL GEOGRAPHY AND FOOD SECURITY	4	3	1	0	Class 12th	NIL

Learning Objectives:

- To understand the nature and scope of agricultural geography.
- To provide a detailed analysis of land use- land cover classification by NRSA.
- To enable the learners to appreciate the geographical factors affecting agriculture
- To enable the learner to identify and understand modern agricultural practices.
- To enable the learner to identify and understand the concept and dimensions of food security.

Learning Outcomes:

- A detailed insight into the subfield of agricultural geography.
- An in-depth knowledge of geographical factors affecting agriculture.
- An understanding of models and regionalization of agriculture.
- Knowledge of concepts and dimensions of food security.
- An understanding of challenges, programme and policies related to sustainable agriculture.

Course Outline

UNIT 1: Concept of Agricultural Geography: (7 hrs)

- Nature and Scope, concept and classification of land use- land cover (twenty two fold NRSA).

Unit 2: Geographical Factors affecting Agriculture: (10 hrs)

- Physical, Economic, Technological, Institutional and socio-cultural.

Unit 3: Models, Theories and Regionalization: (10 hrs)

- Whittlesey's classification of Agricultural regions; Agro ecological regions of India

Unit 4: Agricultural Development: (11 hrs)

- Concept and relevance of Sustainable Agriculture, Modern Agricultural Practices (Green Revolution, Organic farming, Precision Agriculture: role of Remote Sensing and GIS modelling, role of Artificial Intelligence)

Unit 5: Food Security: (7 hrs)

- Concept and dimensions, Food security in India: Challenges, Programmes and Policy.

Readings:

- Gregor, H.P.: Geography of Agriculture. Prentice Hall, New York, 1970.
- Grigg, D. (1984): 'An Introduction to Agricultural Geography', Hutchinson Publication, London
- Hussain, M., 2000, Agricultural Geography, Rawat Publications
- Modgal, Suresh, 2017, Food Security of India, National Book Trust, 81-237-7131-2
- Ramaswamy, S. and Surulivel, L., 2017, Food Security in India, MJP Publishers, ISBN: 9788180943386, 8180943380
- Singh, J. and Dhillon, S.S. (1988), "Agricultural Geography", 2nd edition, Tata McGraw-Hill, New Delhi
- Swaminathan, M.S., 2016, Combating Hunger and Achieving Food Security, Cambridge University Press, 9781107123113
- Symons, L. (1972): 'Agricultural Geography', Bell and Sons, London.
- Tarrant, J.R.(1974): Agricultural Geography, Problems in Modern Geography Series, John Wiley and Sons
- माजिद हुसैन, 2000, कृषि भूगोल, Rawat Publications, 9788170335658

DISCIPLINE SPECIFIC CORE COURSE – RESEARCH METHODOLOGY AND FIELDWORK (PRACTICAL) (DSC 15)

Course title & Code	Credits	Duration (Hrs per week)			Eligibility Criteria	Prerequisite
		Lecture	Tutorial	Practical/ Practice		
RESEARCH METHODOLOGY AND FIELDWORK (PRACTICAL)	4	2	0	2	Class 12th	NIL

Learning Objectives:

1. To form an understanding of various dimensions of fieldwork and its role in geographical studies.
2. To understand in detail various field techniques .
3. Understanding of nuances of research instruments, field tools and report writing.

UNIVERSITY OF DELHI

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Dated: 07.08.2024

NOTIFICATION

Sub: Amendment to Ordinance V

{ECR No. 5-22/ dated 27.07.2024}

Following addition be made to Appendix-II-A to the Ordinance V (2-A) of the Ordinances of the University;

Add the following:

The modifications in DSC Paper titled Research Methodology and Field Work offered as DSC in Semester-V by Department of Geography as under:

Existing	Proposed
Semester-V DSC-15: Research Methodology and Field Work (Practical) (L-T-P=2-0-2)	Semester-V DSC-15: Fieldwork and Field Techniques in Geography (Practical) (L-T-P=0-0-4)

Comparison of Existing and Proposed Course Content

Title & Units	Existing	Proposed
	DSC 15: Research Methodology and Fieldwork (PRACTICAL)	DSC 15 : Fieldwork and Field Techniques in Geography (PRACTICAL)
Unit 1	Research methodology and fieldwork: concept, relevance, ethics and steps.	Pre Field: Doing Fieldwork: Understanding spatial associations in the field and delineating study area; Pilot surveys in Human and Physical Geography.
Unit 2	Framing a research proposal: Identifying the research problem and study area, literature review, research questions, hypothesis, objectives, delineating the database and methods, framing the study relevance.	In- Field: Defining Relevance: Framing the relevance of the study based on spatial associations and field observations, prior studies, narratives.
Unit 3	Methods of Data collection and fieldwork: Observation, Questionnaires, Interviews, Transects and Quadrants, Triangulation, pilot surveys, Recent trends	In- Field: Data collection, tools of fieldwork and Ethics: Collection of field data, through Observation, Questionnaires, Interviews, Transects and Quadrants, Triangulation, informed consent, ethics in the field.
Unit 4	Data analysis and interpretation: Qualitative and Quantitative techniques of analysis; interpreting research findings	Post Field: Analyzing and Interpreting Field Data: Laboratory based data processing, analysis and interpretation, cartographic representations.
Unit 5	Field Report: Organisation and preparation, referencing endnote, footnotes, supplementary materials.	Post- Field: Preparation of field report: Interpreting, organizing and reporting field data, preparation of project report on either a field-based case study outside the NCR or be based on field work within NCR.

Neel Kumar
REGISTRAR